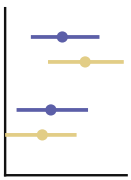
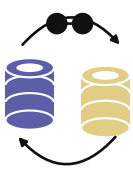
	Indicators	Interpretations
Fidelity	*How well synthetic data resemble real data*	
	Univariate analysis • Bar plots (categorical) or density plots (continuous) • Jensen–Shannon divergence (JSD) per variable	• Visual comparison of marginal distributions • Lower JSD indicates better marginal fidelity
	Bivariate analysis • Pearson correlations (continuous) or Cramér's V (categorical) • Correlation Frobenius distance (CFD) between correlation matrices	• Preservation of pairwise dependencies • Lower CFD indicates better fidelity
	Multivariate analysis • Logistic regression distinguishing real from synthetic records (DiscPred); mean AUC, 5-fold CV	• DiscPred closer to 0.5 indicates better fidelity
Utility	*Ability of synthetic data to substitute for real data in analyses and predictive tasks*	
	**Analytical utility** • Multivariable regression (logistic: PIMA; Cox: AIDS, REIN) on real and synthetic data • Standardized difference (SDiff) of OR / HR estimates: mean [q0.05–q0.95]	• Agreement of inferences from synthetic and real data • Lower SDiff indicates better analytical utility
	Predictive utility • TRTR (reference), TSTR and TSRTR scenarios, evaluated on the real test set • AUC (PIMA) or time-dependent AUC (AIDS, REIN): |TSTR − TRTR| and |TSRTR − TRTR|	• Values closer to 0 indicate that models trained with synthetic data perform like models trained on real data, i.e., higher predictive utility
Privacy	*Protection of individuals in the real data against disclosure of personal information*	
	**Record-level proximity** • Record matching (RM): mean Gower distance from real training records to their closest synthetic record • Distance to the closest record (DCR): frequency of synthetic records below the real-data 2nd-percentile threshold • Nearest neighbor distance ratio (NNDR): 1st / 2nd closest real neighbor; 5th percentile	• Higher RM indicates lower record-level similarity • DCR ≈ 1: as in real data; well above 1: excess of very close records • Lower NNDR: closest real record much nearer than the second
	Membership inference risk • Holdout set distinguishing: proximity of synthetic records to training vs. holdout records, $MIR_{HoldoutSet}$	• Close to 0.5: stronger membership privacy; well above 0.5: potential leakage
	Attribute inference risk • 1-nearest-neighbor attack (Gower) over six auxiliary-information scenarios; mean F1-score, AIR_{F1}	• Lower AIR_{F1} indicates lower attribute inference risk
	Identity disclosure risk • Linkability attack: same synthetic record retrieved from two disjoint attribute subsets; success rates r_{train}, $r_{control}$ (records unused for generation) and r_{naive} (random matching) • Attack strength: $s = r_{train} - r_{naive}$ • Specific linkability risk: $R = (r_{train} - r_{control})/(1 - r_{control})$	• $s > 0$: attack outperforms the naive baseline; otherwise not informative for risk assessment • R close to 0: similar success on training and control records, limited evidence of leakage; larger positive R: stronger signal of potential information leakage